

Rigid Analytic Geometry Week 6 Solution

So Murata

SoSe 26, University of Bonn

Problem 1

Set

$$\xi_R := \{\mathcal{R}(f_0|f_1, \dots, f_m) \mid N_R(f_j) \leq N_R(f_0) \forall j\}.$$

We first show that this is a vander Put point. Take,

$$\Omega = \mathcal{R}(f_0|f_1, \dots, f_m), \Omega' = \mathcal{R}(g_0|g_1, \dots, g_n) \in \xi_R.$$

Then by the multiplicativeness of N_R , we see that

$$N_R(f_i g_j) = N_R(f_i) N_R(g_j) \leq N_R(f_0 g_0).$$

Let Ω' be a rational open such that for some $\Omega \in \xi_R$, we have $\Omega \subseteq \Omega'$. Then write,

$$\Omega' = \mathcal{R}(g_0 \mid g_1, \dots, g_n).$$

Let $h_i = g_0/g_i$, then as $\Omega \subseteq \Omega'$, we have Therefore ξ_R is a van der Put point. This satisfies both conditions. Since the membership is explicitly given, clearly this is a unique van der Put point satisfying two given conditions.

Problem 2

By assumption, there is $a \in K^\times$ such that $|a| = R$. Consider,

$$\Omega = \mathcal{R}(a|T_1 - x).$$

Then for any $z \in \Omega$, we have

$$|z - x| \leq R.$$

Therefore, $\Omega = B_R(x)$.

Problem 3

Note that

$$K_\rho(0) \in \xi_R \iff N_R(T) < N_R(\rho).$$

Let $\Omega = \mathcal{R}(f_0|f_1, \dots, f_m) \in v, \Omega \notin \xi_R$, then there is j such that

$$N_R(f_j) > N_R(f_0).$$

Let $g = \frac{f_j}{f_0}$, then there is a coefficient a_n of g of degree n , such that

$$|a_n| R^n > 1.$$

Using the assumption,

$$|a_n|\rho^n > 1.$$

In particular $N_R(g) > 1$. Therefore $K_\rho(0) \not\subseteq \Omega$. By the equivalent condition for $K_\rho(0) \in v$ and that it is a filter, we have

$$\Omega \cap K_\rho(0) \in v.$$